Space Weather Events – Background Information and Additional Resources

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Background information:

- The Earth has a **magnetic field** that is mostly generated by the movement of liquid iron in the Earth's outer core (NASA, 2021). The magnetic field extends out from Earth into space and forms the **magnetosphere,** which shields the planet from harmful solar particles (Olsen & Pauluhn, 2019) (see **Figure 1**).

- **Space weather events** are events driven by distant solar activity that impact human technology and the near-Earth space environment (BoM SWS).

- The three main drivers of space weather events are (NOAA SWPC):
  - Solar flares: large eruptions of electromagnetic radiation from the Sun.
  - Coronal mass ejections (CMEs): expulsions of plasma and magnetic field from the Sun's corona (outermost layer).
  - High speed solar wind streams: solar wind of up to 500-800 km/s produced from coronal holes.

- **Geomagnetic storms** are major disturbances in Earth's magnetosphere that result from energy exchange between the solar wind and the near-Earth space environment (NOAA SWPC).

- **CMEs** are responsible for the most extreme geomagnetic storms. They travel from the Sun at speeds ranging from **< 250 km/s** to up to **3000 km/s**. Fast CMEs can reach Earth in **under 24 hours**, whilst slower ejections can take **several days** to arrive (NOAA SWPC).

- Space weather can produce **hazardous conditions** on Earth. It can disrupt technology and infrastructure that modern society relies on (see **Figure 2**), as well as produce auroras. Space weather hazards include (NOAA SWPC):
  - Disruption to electrical power grids
  - Loss of HF radio communications
  - Loss of satellite communications
  - Satellite drag
  - Errors in GPS positioning

Below are some example extreme space weather events:

- The "**Carrington Storm**" on the 2nd of September 1859 was one of the most intense geomagnetic storms ever recorded. It is the archetype of an extreme space weather event and disrupted global telegraph communication (Love et al., 2025). If an event of this scale were to occur today, it could cause trillions of dollars' worth of damage and take years to recover from.

- The **March 1940 geomagnetic superstorm** severely disrupted global communication and shut down trans-Atlantic telegraph, telephone and radio networks. It also disrupted communications in Sydney and was reported as the worst magnetic storm in modern history (The Sydney Morning Herald, 1940).

- In February 2022, **Starlink** (a SpaceX global satellite internet network) launched 49 satellites into an already disturbed space environment and lost 38 of these. The satellite loss was due to increased atmospheric drag resulting from geomagnetic storm conditions, which was poorly modelled (Fang et al., 2022).

- Space weather events and their associated hazards are communicated to the public using **space weather scales**. There is no internationally agreed-upon scale, however geomagnetic storms are generally classified on a 1-5 system from G1 (minor) to G5 (extreme).
  - The Australian Space Weather Alert System is available here
  - The NOAA space weather scale which is used in this class is available here (see **Figure 3**)

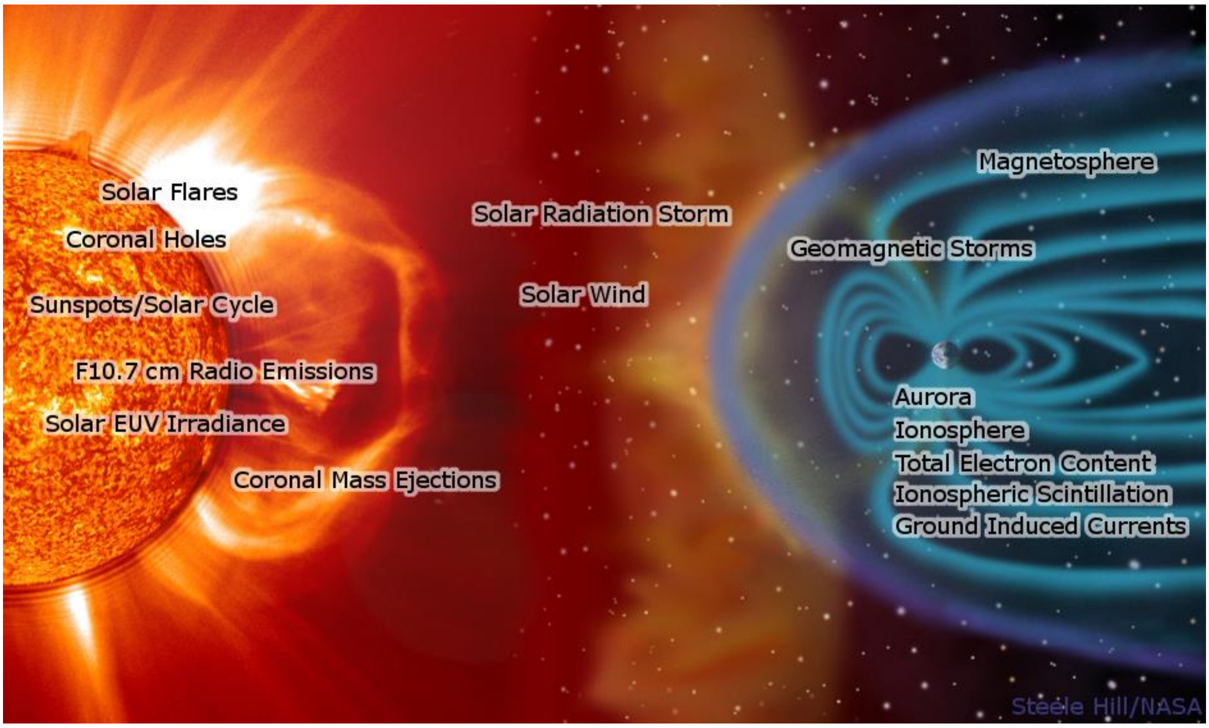

Figure 1: The Earth's magnetosphere shields the planet from harmful solar particles. The main features on the Sun and the Earth related to space weather are highlighted (European Space Agency - ESA).

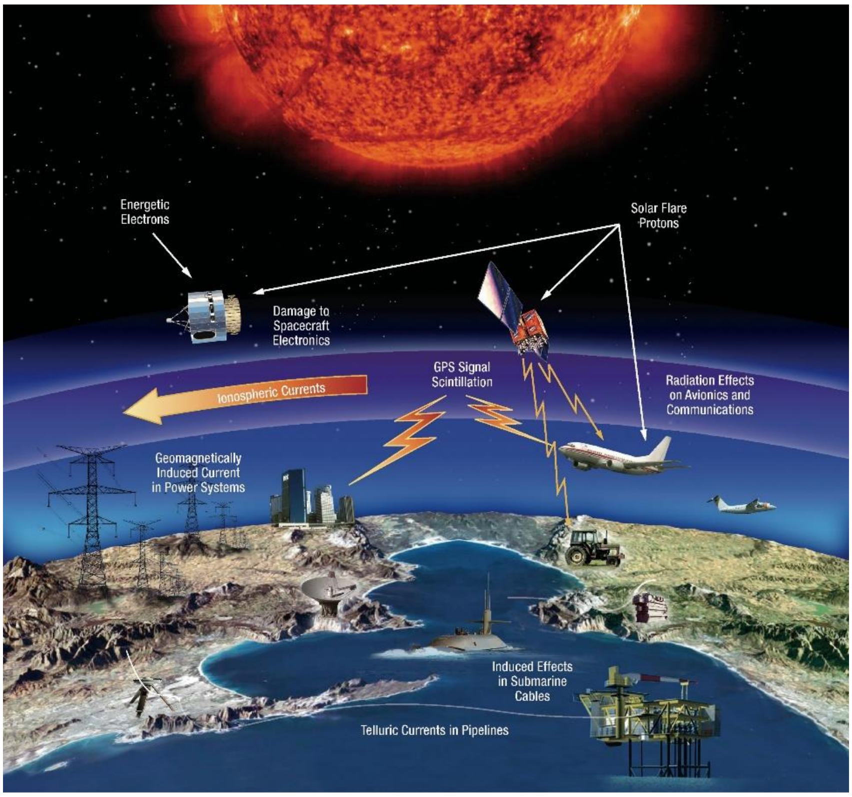

Figure 2: Some of the impacts of space weather on human technology and infrastructure (NASA's Goddard Space Flight Center).

NOAA Space Weather Scales

Category		Effect	Physical measure	Average Frequency (1 cycle = 11 years)
Scale	Descriptor	Duration of event will influence severity of effects		
Geomagnetic Storms			Kp values* determined every 3 hours	Number of storm events when Kp level was met; (number of storm days)
G 5	Extreme	Power systems: widespread voltage control problems and protective system problems can occur, some grid systems may experience complete collapse or blackouts. Transformers may experience damage. Spacecraft operations: may experience extensive surface charging, problems with orientation, uplink/downlink and tracking satellites. Other systems: pipeline currents can reach hundreds of amps, HF (high frequency) radio propagation may be impossible in many areas for one to two days, satellite navigation may be degraded for days, low-frequency radio navigation can be out for hours, and aurora has been seen as low as Florida and southern Texas (typically 40° geomagnetic lat.).**	Kp=9	4 per cycle (4 days per cycle)
G 4	Severe	Power systems: possible widespread voltage control problems and some protective systems will mistakenly trip out key assets from the grid. Spacecraft operations: may experience surface charging and tracking problems, corrections may be needed for orientation problems. Other systems: induced pipeline currents affect preventive measures, HF radio propagation sporadic, satellite navigation degraded for hours, low-frequency radio navigation disrupted, and aurora has been seen as low as Alabama and northern California (typically 45° geomagnetic lat.).**	Kp=8	100 per cycle (60 days per cycle)
G 3	Strong	Power systems: voltage corrections may be required, false alarms triggered on some protection devices. Spacecraft operations: surface charging may occur on satellite components, drag may increase on low-Earth-orbit satellites, and corrections may be needed for orientation problems. Other systems: intermittent satellite navigation and low-frequency radio navigation problems may occur, HF radio may be intermittent, and aurora has been seen as low as Illinois and Oregon (typically 50° geomagnetic lat.).**	Kp=7	200 per cycle (130 days per cycle)
G 2	Moderate	Power systems: high-latitude power systems may experience voltage alarms, long-duration storms may cause transformer damage. Spacecraft operations: corrective actions to orientation may be required by ground control; possible changes in drag affect orbit predictions. Other systems: HF radio propagation can fade at higher latitudes, and aurora has been seen as low as New York and Idaho (typically 55° geomagnetic lat.).**	Kp=6	600 per cycle (360 days per cycle)
G 1	Minor	Power systems: weak power grid fluctuations can occur. Spacecraft operations: minor impact on satellite operations possible. Other systems: migratory animals are affected at this and higher levels; aurora is commonly visible at high latitudes (northern Michigan and Maine).**	Kp=5	1700 per cycle (900 days per cycle)

* Based on this measure, but other physical measures are also considered.
** For specific locations around the globe, use geomagnetic latitude to determine likely sightings (see www.swpc.noaa.gov/Aurora)

Figure 3: NOAA Space Weather Scale for Geomagnetic Storms, G1 (minor) – G5 (extreme) (NOAA/ SWPC).

Some of this content was inspired by a Space Weather Hazard Communication class developed by Associate Professor Maria Seton, University of Sydney.

References:

Bureau of Meteorology Australian Space Weather Forecasting Centre (BoM SWS). *What is Space Weather?* Available at: https://www.sws.bom.gov.au/Educational/1/2

Fang, T.-W., Kubaryk, A., Goldstein, D., Li, Z., Fuller-Rowell, T., Millward, G., et al. (2022). Space weather environment during the SpaceX Starlink satellite loss in February 2022. *Space Weather*, 20, e2022SW003193. https://doi.org/10.1029/2022SW003193

Love, J. J., Lucas, G. M., Kelbert, A., Rigler, E. J., Bedrosian, P. A., & Schnepf, N. R. (2025). Mapping a Carrington storm. *Geophysical Research Letters*, 52, e2025GL116835. https://doi.org/10.1029/2025GL116835

NASA (2021). *Earth's Magnetosphere: Protecting Our Planet from Harmful Space Energy.* Available at: https://science.nasa.gov/science-research/earth-science/earths-magnetosphere-protecting-our-planet-from-harmful-space-energy/

National Oceanic and Atmospheric Administration (NOAA) Space Weather Prediction Center (SWPC). *Coronal Mass Ejections*. Available at: https://www.spaceweather.gov/phenomena/coronal-mass-ejections

National Oceanic and Atmospheric Administration (NOAA) Space Weather Prediction Center (SWPC). *Geomagnetic Storms*. Available at: https://www.spaceweather.gov/phenomena/geomagnetic-storms

National Oceanic and Atmospheric Administration (NOAA) Space Weather Prediction Center (SWPC). *Solar Flares (Radio Blackouts).* Available at: https://www.spaceweather.gov/phenomena/solar-flares-radio-blackouts

National Oceanic and Atmospheric Administration (NOAA) Space Weather Prediction Center (SWPC). *Solar Wind*. Available at: https://www.spaceweather.gov/phenomena/solar-wind

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National Oceanic and Atmospheric Administration (NOAA) Space Weather Prediction Center (SWPC). *NOAA Space Weather Scales.* Available at: https://www.spaceweather.gov/noaa-scales-explanation

Olsen, N. and Pauluhn, A. (2019). Exploring Earth's magnetic field – Three make a Swarm. *Spatium*, 2019(43), pp. 3-15. Available at: https://backend.orbit.dtu.dk/ws/portalfiles/portal/185293289/Exploring_Earths_magnetic_field.pdf

The Bureau of Meteorology (2024). *Australian Space Weather Alert System.* Available at: https://www.sws.bom.gov.au/Category/About%20SWS/Australian%20Space%20Weather%20Alert%20System/BoM%20Australian%20Space%20Weather%20Alert%20System.pdf

The Sydney Morning Herald (1940), 'Big magnetic storm', 26 March, p. 9. Available at: https://trove.nla.gov.au/newspaper/article/17663722/1101773

Additional resources:

Further coronagraph images, videos and near real-time information about the Sun can be found on the Solar and Heliospheric Observatory (SOHO) website: National Aeronautics and Space Administration (NASA) SOHO – Solar and Heliospheric Observatory. Available at: https://soho.nascom.nasa.gov/

Further information on the impacts of space weather:

National Oceanic and Atmospheric Administration (NOAA) Space Weather Prediction Center (SWPC). *Space Weather Impacts.* Available at: https://www.spaceweather.gov/impacts

A NASA YouTube video that explains the physics of the magnetosphere, how geomagnetic storms form, their impacts on Earth, and the satellite missions used to study them: NASA Science (2018). *NASA ScienceCasts: Earth's Magnetosphere*. Available at: https://youtu.be/o4FSg-90XlA?si=5Q8FNLZNiqVHR-ld

A timeline of major space weather events is contained within the BoM Australian Space Weather Alert System Brochure: Australian Government Bureau of Meteorology (2024). *Australian Space Weather Alert System.* Available at: https://www.sws.bom.gov.au/Category/About%20SWS/Australian%20Space%20Weather%20Alert%20System/BoM%20Australian%20Space%20Weather%20Alert%20System.pdf

Current geomagnetic conditions can be found on the following sites:

Bureau of Meteorology Space Weather Services (BoM SWS). *Space Weather Conditions*. Available at: https://www.sws.bom.gov.au/

National Oceanic and Atmospheric Administration (NOAA), Space Weather Prediction Center (SWPC) (2015). *3-Day Geomagnetic Forecast.* Available at: https://www.spaceweather.gov/products/3-day-geomagnetic-forecast

A NASA guide to studying space weather with a smartphone magnetometer: NASA (2025). *Smartphone Magnetometers for Space Weather Studies.* Available at: https://science.nasa.gov/learn/heat/resource/smartphone-magnetometers-student-guide/